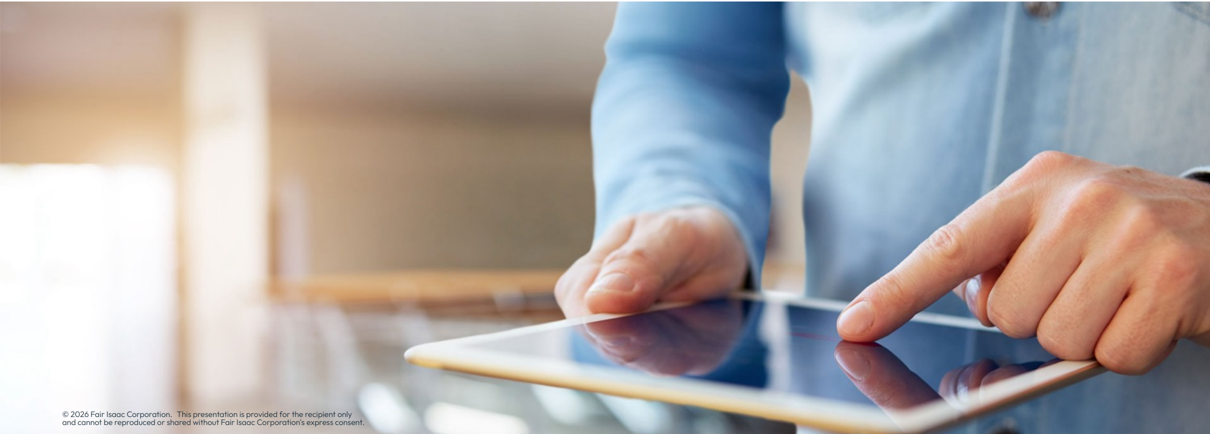
Nonlinear



Modeling nonlinear problems

Modeling nonlinear problems in Python

- Nonlinear problems, i.e. problems containing at least one nonlinear constraint or objective, can be modeled via the Xpress Python interface:
  - Nonlinear expressions follow the same relational and arithmetic logic as linear expressions
  - Available arithmetic operators: `+`,`–`, `*`, `/`, `**` (which is the Python equivalent for the power operator, “^“)
  - Univariate functions can be used from the following list: `sin`, `cos`, `tan`, `asin`, `acos`, `atan`, `exp`, `log`, `log10`,`abs`, `sign`, and `sqrt`
  - The multivariate functions `min` and `max` can receive an arbitrary number of arguments

Modeling nonlinear problems in Python

- Examples of nonlinear problem elements:

```
p.addConstraint(x**4 + 2 * x**2 - 5  >= 0)   # polynomial constraint
p.addConstraint(xp.sin(math.pi * x) == 0)    # terrible way to constrain x to be integer
p.addConstraint(x**2 * xp.sign(x) <= 4)      # signum function
p.setObjective((a-x)**2 + b*(y-x**2)**2)     # minimize Rosenbrock function
```

Finding help: *For more information about modeling nonlinear problems, browse the FICO® Xpress NonLinear reference manual*

User functions

- A user function enables the creation of an expression that is computed through external code:
  - Any user-defined function can be called within a problem by using the function xpress.user():

```
xp.user(f, a1, a2, ...)
```

    - Where `f` represents the user-defined function name and `a1, a2, ...` the necessary arguments, as in the example below:

```
def myfunc(v1, v2, data):
    model = MLmodel(v1, v2, data)    # MLmodel() defined elsewhere
    return model.results

  data = readData()  # readData() defined elsewhere
  x, y = p.addVariable(), p.addVariable()
  p.setObjective(xp.user(myfunc, x, y, data))
```

- You can define user functions with a simulation or machine learning model!
  - Be aware of losses in determinism and performance
- User functions are not supported by FICO® Xpress Global

Thank You

www.fico.com/optimization

FICO